

Information Retrieval

Course Metadata

Credit Structure	L-T-P: 3-0-0 Total: 3
Offering Semester	5 th , Autumn
Level of Subject	UG Level
Compulsory or elective?	Core
Pre-requisite	NIL
Faculty Names	Koustav Rudra, Jiaul H Paik
Overlapping Subjects	Information Retrieval (CS60080/CS60092) $\approx$ 80%
Justification of overlap	This course will be a core subject for BTech in AI program

Course Description

In the digital era we are sitting over large volume of information. However, information is nothing without retrieval. Information come in the form of both structured and unstructured. There exist several steps in processing, storing, and retrieving information. The storing step involves document indexing, index optimization, retrieval step includes query, document representations, vector space mappings, ranking metrics, semantic representations, etc. Machine learning, like in many other domains, have been used to automate and improve many of these tasks such dense embedding, neural information retrieval, etc. A new paradigm of information retrieval (neural information retrieval) has emerged and it has been widely adopted in recent search techniques such as document retrieval, question answering, conversational search, etc. Deep learning helps in significantly improving the retrieval and ranking performance. Apart from that recently introduced large language models (ChatGPT, LLAMA, etc.) helps in understanding query intentions and improve the ranking performance through prompt and in-context learning.

Considering the rapid adoption of neural information retrieval and its effectiveness, the aim of this course is to introduce students to basic and some advanced, recent concepts of information retrieval and their use in real world applications in diverse domains.

Course Content

- Introduction and application of information retrieval
- Basics of indexing: Document preprocessing, Index construction, Index compression
- Vector space model, Relevance feedback, Evaluation of retrieval methods
- Retrieval from semi-structured documents
- Probabilistic information retrieval and Language models
- Latent semantic indexing and Link analysis
- Dense indexing and LLM based retrieval

Lecture-wise Breakup

Sl No.	Module	Unit	Module Hr (39)
1	Introduction	Introduction to information retrieval	1
		Applications of information retrieval	
2	Boolean and Tolerant Retrieval	Inverted index	5
		Document processing	
		Posting list, skip pointer and phrase query	
		Wildcard queries and correction	
3	Index Construction and Compression	Block sort based, Single-pass in-memory, Distributed, Dynamic indexing	5
		Dictionary compression	
		Posting file compression	
4	Scoring and Evaluation	Term frequency and weighting, Vector space model, Variant tfidf functions	5
		Efficient scoring and ranking: Inexact top K retrieval, Champion list, Cluster pruning	
		Evaluation: Ranked and unranked retrieval sets, Assessing relevance	
5	Relevance Feedback and Query Expansion	Relevance feedback: Rocchio algorithm, Probabilistic relevance feedback, Pseudo relevance	3
		Query reformulation and expansion	
6	Retrieval from semi-structured document	Field based retrieval in semi-structured documents	2
7	Probabilistic and Language Model for Information Retrieval	Probabilistic model: Probability Ranking Principle, Binary Independence Model	4
		Language models, Query likelihood model	
8	Text Classification and Clustering	Naïve Bayes, Bernoulli model for text classification	5
		Vector space classification: Rocchio classification, K nearest neighbour	
		Clustering: Flat and hierarchical clustering	
9	Latent Semantic Indexing	Term-document matrix, Singular value decomposition, Low rank approximation	3
10	Link Analysis	Web as graph, PageRank, Hubs and Authorities	2
11	Dense Indexing	Bi-encoder for dense retrieval (DPR, ANCE), Late interaction (ColBERT)	2
12	LLM and Retrieval	Applications of Large language models in query reformulation, document retrieval	2

Books

1. **An Introduction to Information Retrieval**, Christopher D. Manning, Prabhakar Raghavan, Hinrich Schütze, Cambridge University Press, 2009
2. **Pretrained Transformers for Text Ranking: BERT and Beyond (Synthesis Lectures on Human Language Technologies)**, Jimmy Lin, Rodrigo Nogueira, Andrew Yates, Morgan & Claypool Publishers, 2021
3. **Question Answering for the Curated Web: Tasks and Methods in QA over Knowledge Bases and Text Collections**, Rishiraj Saha Roy, Avishek Anand, Synthesis Lectures on Information Concepts, Retrieval, and Services, Springer, 2021

References

1. SC. Lin, JH. Yang, J. Lin. "Distilling dense representations for ranking using tightly-coupled teachers". arXiv preprint arXiv:2010.11386. 2020 Oct 22
2. O. Khattab, M. Zaharia. "Colbert: Efficient and effective passage search via contextualized late interaction over bert". In Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval 2020 Jul 25 (pp. 39-48)
3. L. Xiong, C. Xiong, Y. Li, KF. Tang, J. Liu, P. Bennett, J. Ahmed, A. Overwijk. "Approximate nearest neighbor negative contrastive learning for dense text retrieval". arXiv preprint arXiv:2007.00808. 2020 Jul 1
4. Y. Zhu, H. Yuan, S. Wang, J. Liu, W. Liu, C. Deng, Z. Dou, JR. Wen. "Large language models for information retrieval: A survey". arXiv preprint arXiv:2308.07107. 2023 Aug 14
5. Q. Ai, T. Bai, Z. Cao, Y. Chang, J. Chen, Z. Chen, Z. Cheng, S. Dong, Z. Dou, F. Feng, S. Gao. "Information retrieval meets large language models: a strategic report from chinese ir community". AI Open. 2023 Jan 1;4:80-90